

EPHRAIM KHALAUHENI MANDALA

Biomedical Engineer

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Malawian | Male
English, Chichewa, Tonga, Tumbuka
MEI Registered | ABEPM Member

PROFESSIONAL SUMMARY

I am highly motivated and results-driven Biomedical Engineer with a strong track record in medical equipment maintenance, quality assurance, clinical research and electrical systems. I have demonstrated success in laboratory diagnostics, equipment calibration, electrical safety testing, SOP development and clinical staff training. I am Committed to enhancing healthcare delivery through innovation, system optimization and regulatory compliance, with growing expertise in data science, project management and sales strategy.

EDUCATION & CERTIFICATIONS

Academic Qualifications

BSc (Hons) in Biomedical Engineering — Malawi University of Business and Applied Sciences (MUBAS) (2023)

Diploma in Sales and Marketing Operations — Uniathena [Pending]

Malawi School Certificate of Education (MSCE) — St. John Bosco Secondary School (2016)

Junior Certificate of Education (JCE) — St. John Bosco Secondary School (2014)

Primary School Leaving Certificate (PSLCE) — Masasa Primary School (2012)

Professional Certifications & Training

Python in Data Science and Machine Learning — Uniathena (2026)

Electrical Safety Certificate — Uniathena (2026)

Electrical Safety and Legal Compliance — Uniathena (2026)

Engineering Management Certificate — Uniathena (2026)

Project Monitoring and Evaluation Certificate — Green Cedar Consult (2024)

ICH-GCP E6(R2) — Global Health Network (2022)

Introduction to Clinical Research & Informed Consent — Global Health Network (2022)

Ethics and Human Subject Protection — Association of Clinical Research Professionals (ACRP) (2022)

WORK EXPERIENCE

Site Biomedical Engineer

Training and Research Unit of Excellence (IMPALA PROJECT) | Zomba | July 2022 – February 2023

- Ensured clinical data integrity and patient safety by conducting rigorous quality control on patient monitors using electrical safety analyzers and ProSim 8 simulators, achieving full compliance with biomedical standards.
- Developed comprehensive logs and SOPs and successfully trained nursing staff on patient monitor usability, improving protocol adherence across the unit.
- Reduced equipment downtime and extended device lifespan by performing scheduled preventive and corrective maintenance on a wide range of laboratory equipment including Sysmex XP-300, HemoCue, slit microscopes, biochemistry analysers, heaters, thermometers, and MRI units.
- Safeguarded research continuity by managing server data backups and ensuring uninterrupted system functionality throughout the study period.
- Strengthened research outcomes by collaborating with nurses to refine usability protocols and streamline clinical workflows.
- Contributed to accurate multilingual documentation by transcribing study nurse audio recordings from Chichewa to English.

Biomedical Engineering Intern

- ▶ Enhanced global biomedical workforce capacity by producing and editing high-quality training and maintenance videos for various medical devices, used across multiple African countries.
- ▶ Expanded the reach of biomedical knowledge by facilitating online webinars attended by engineers and clinicians from across the region.
- ▶ Accelerated digital health initiatives by supporting the BME Konza app and contributing to field research and data analysis activities.
- ▶ Ensured technical accuracy by developing device scripts in full compliance with manufacturer specifications.
- ▶ Supported evidence-based program planning by conducting situational analyses of medical equipment across multiple African countries.

Biomedical Engineering Intern

Bwaila District Hospital | Lilongwe | October 2024-November 2025

- ▶ Improved facility energy resilience by successfully installing solar backup systems that provided reliable power for critical medical equipment.
- ▶ Established a robust equipment management framework by maintaining detailed documentation and updating the hospital's medical device inventory.
- ▶ Standardized maintenance procedures by developing comprehensive SOPs, directly reducing equipment-related incidents.
- ▶ Restored equipment functionality and minimized service interruptions by resolving complex electrical and electronic faults across departments.
- ▶ Delivered infrastructure improvements by leading construction projects at the main hospital and affiliated community health facilities.

Biomedical Technician Intern

Embangweni Mission Hospital | Mzimba | 2021-2022

- ▶ Improved living conditions for 40 staff members by successfully installing water supply submersible pumps and solar systems, delivering reliable water and energy access.
- ▶ Strengthened asset management by accurately recording and organizing the hospital's medical device inventory.
- ▶ Maintained operational readiness of over 100 medical devices by conducting systematic preventive maintenance and targeted troubleshooting.
- ▶ Achieved a 60% reduction in device downtime by spearheading the development and implementation of a Planned Asset Maintenance (PAM) system and standardized SOPs.

TECHNICAL SKILLS

Programming	C++, Python (Data Science & ML), HTML, CSS, JavaScript, MATLAB/Simulink
Data & Research	ODK, KoboToolbox, REDCap, SPSS, STATA, NVivo, ATLAS.ti, Microsoft Office Suite
Hardware & Systems	Digital/Analog Circuit Design, Microcontrollers, Embedded Systems, Electrical Safety Testing, Signal Analysis & System Calibration
Instrumentation	Analyzers, Multimeters, Spectrophotometers, Thermometers, Microscopes, Sensors & Instrumentation Integration
Compliance & QA	Regulatory Compliance Testing, SOP Development, Electrical Safety & Legal Compliance, Engineering Management

KEY PROJECTS & ACHIEVEMENTS

- ▶ Led quality assurance testing of Impala Patient Monitors using ProSim 8 and electrical safety analyzers, ensuring clinical-grade device performance.
- ▶ Contributed to an HDU Visualisation Framework project that improved early vital sign interpretation for clinical decision-making.
- ▶ Designed and supervised COVID-19 patient isolation tents at Queen Elizabeth Central Hospital under the Malawi Engineering Institution (MEI).
- ▶ Delivered a 60% reduction in device downtime at Embangweni Mission Hospital through the establishment of a proactive Planned Asset Maintenance (PAM) programme.

REFERENCES

Mr. Isaac Shawa

Electrician & Biomedical Technician
Embangweni Mission Hospital, Mzimba
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Mr. Lyd Kayembe

Program Manager
Oxygen Alliance, GHII, Lilongwe
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Miss Sheira Banda

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